

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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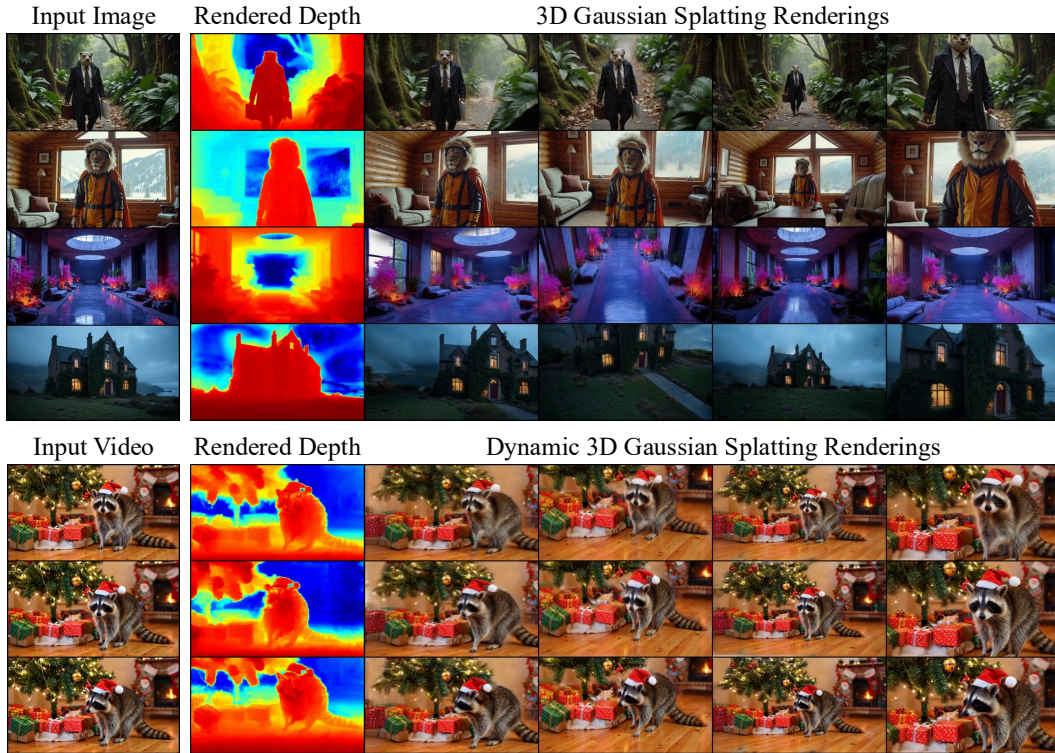


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward manner by self-distilling a video diffusion model, without requiring real-world multi-view data. From a video input (bottom), *Lyra* infers a dynamic 3DGS representation that supports control over both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is a broadly useful capability for interactive media, simulation, and physical AI settings such as robotics, autonomous driving, and industrial automation. Learning-based 3D reconstruction systems typically depend on captured real-world multi-view data, a requirement that limits deployment in domains where such data are costly, sparse, or unavailable. Video diffusion models provide a complementary source of visual and geometric priors: they can synthesize diverse environments and camera motions, but their 2D frame outputs do not directly support applications that require navigation, interaction, or view-consistent simulation. We propose a self-distillation framework that transfers the implicit 3D knowledge of a camera-controlled video diffusion model into an explicit 3D Gaussian Splatting (3DGS) representation, avoiding the need for captured multi-view training data. Concretely, we augment the standard RGB decoder with a 3DGS decoder and supervise it using the RGB output of the same video model. This design enables the 3DGS decoder to be trained entirely from synthetic supervision generated by video diffusion models while preserving the

original datasets, tasks, and camera-control assumptions of the study. At inference time, the resulting model synthesizes 3D scenes from either a text prompt or a single image and supports real-time rendering. The same framework also extends to dynamic 3D scene generation from a monocular input video. Across the static and dynamic benchmarks considered here, our framework achieves state-of-the-art performance for generative 3D and 4D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale remains a central challenge for computer vision and graphics, with impact across film-making, VR/AR, embodied AI, and closed-loop simulation. Across these application areas, explicit 3D representations are valuable because they support real-time rendering, physical interaction, and consistent synthesis from viewpoints beyond the original observation.

Recent neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) can recover such representations from posed images. However, their dependence on accurate camera poses and high-quality multi-view imagery limits how broadly they can be applied. The constraint is even stronger for dynamic scenes, where synchronized multi-camera capture is often required (Li et al., 2022). Reconstruction methods are also tied to observed content and therefore have limited ability to infer plausible unobserved regions. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce some per-scene optimization costs, but they inherit a different bottleneck: diverse, large-scale 3D training data are scarce, which weakens generalization beyond the data distributions on which they are trained.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a broader source of scene priors. Because they are trained on large video corpora, they can synthesize varied environments and camera motions with strong visual fidelity. Through exposure to real-world videos spanning handheld motion, cinematic camera paths, and drone footage, these models implicitly learn cues about 3D structure without requiring explicit multi-view supervision. Unlike reconstruction systems (Charatan et al., 2024), generative video models can also produce plausible content outside the visible input frames. Yet their outputs are still 2D videos rather than explicit 3D assets, which limits their direct use in simulation, long-horizon view synthesis, and tasks requiring geometric consistency or physical interaction. Recent work (Wang et al., 2024e; Ren et al., 2025) shows that video models can be adapted for explicit camera control using relatively modest posed datasets. This makes camera-controlled video generation a useful bridge: it can provide both generated inputs and structured supervision for feed-forward 3D reconstruction.

Motivated by this connection, we combine reconstruction and generation under the framing of *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments from the latent representations of a video diffusion model in a single forward pass. Crucially, *Lyra* is trained through self-distillation (Fig. 2), where a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a camera-controlled video diffusion model. Given a single input image or video, we sample a camera trajectory, denoise the corresponding video latent, and decode that latent through two parallel branches. The standard RGB decoder synthesizes a video sequence, while the 3DGS decoder produces an explicit 3D representation. These branches form a teacher–student loop: the RGB branch acts as the teacher and the 3DGS branch as the student. To broaden viewpoint coverage within each generated scene, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse them while reducing long-range and multi-view inconsistencies.

This generative reconstruction framework provides three benefits that extend beyond a single dataset or scene type within the conditions evaluated here: **(i)** it can synthesize large-scale environment data across diverse prompts directly from video diffusion models, removing the need for captured real-world multi-view datasets; **(ii)** by operating in the latent space of a video model, it processes many generated views more efficiently than pixel-space feed-forward reconstruction; and **(iii)** its explicit 3DGS output provides view-consistent geometry that can be rendered in real time and used in downstream simulation workflows. Consequently, at inference time, *Lyra* generates high-quality 3D Gaussian scenes from monocular inputs without requiring additional optimization or post-processing.

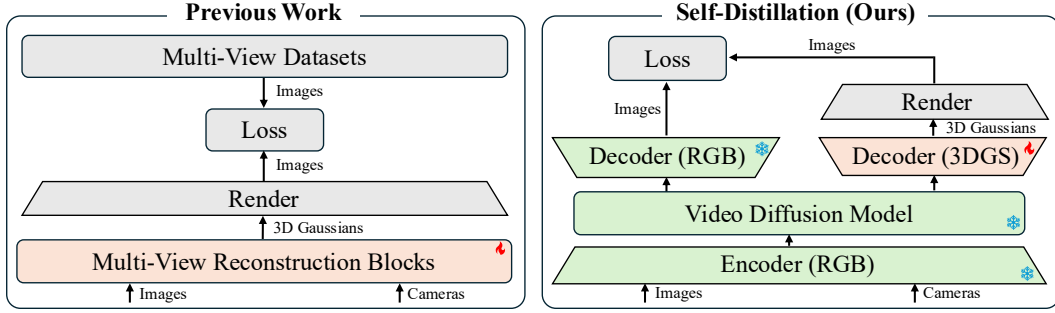


Figure 2: **Self-distillation framework of *Lyra*.** Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). We instead use self-distillation (right) for generative 3D scene reconstruction: a pre-trained camera-controlled video diffusion model and its RGB decoder output serve as the teacher, supervising renderings from the 3DGS decoder student. Because the teacher is pre-trained on diverse 2D video data, it provides broad synthetic multi-view supervision without captured multi-view datasets.

We further adapt the same self-distillation principle to dynamic 4D generation from monocular video. In this setting, the video model provides space-time supervision, and the student learns time-conditioned 3DGS representations for novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- We extend the same framework to dynamic scenes, enabling 4D reconstruction from a monocular video under the study’s video-generation and camera-control setup.
- We show that the resulting model generalizes across diverse evaluated scenes, achieving state-of-the-art results in single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily addressed object-centric settings without complex backgrounds. Later work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) broadened this direction toward scene-scale generation. CAT3D (Gao et al., 2024b) adapts a pre-trained image diffusion model to generate multiple views from either generated or real image inputs. These views are then reconstructed into an explicit 3D representation, such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related methods further extend multi-view generation to dynamic 3D scenes by using space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Although this family of methods demonstrates strong visual results across several generation settings, converting generated views into explicit 3D typically requires a costly optimization stage that is repeated for every scene. Our work instead targets direct 3D scene generation from text or a single image, using generated multi-view video supervision without a separate per-scene reconstruction optimization.

Camera-conditioned video generation. Camera control for video diffusion models has advanced rapidly. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models with camera poses. Follow-up methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) use Plücker coordinates for pixel-wise camera conditioning, while another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) studies training-free camera control. Recent systems such as SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) show that large-scale synthetic data can support compelling camera-controlled video synthesis. These models broaden controllable video generation, but their outputs remain 2D and are not themselves persistent 3D representations. Our work is complementary: we use a camera-controlled video model to train a feed-forward 3D generator whose output can be rendered from new viewpoints after generation. Specifically, we build on GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and use it as the teacher in a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction work largely considered object-centric scenes without backgrounds, predicting NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or images. More recent scene-level methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) typically regress per-pixel 3D Gaussians from one or more images. However, many scene-level reconstruction models remain tied to their training distributions, such as RealEstate10K (Zhou et al., 2018), and therefore generalize less reliably to generated or out-of-distribution scenes. Recent generative feed-forward models attempt to expand this scope. Bolt3D (Szymanowicz et al., 2025b), for example, trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps along with multi-view images for feed-forward 3D reconstruction. Closest to our setting, Wonderland (Liang et al., 2025a) uses a camera-controlled video model and a feed-forward network to generate 3D Gaussians. In contrast, we use the camera-controlled video model as a teacher for self-distillation, training the student 3D model without requiring real-world multi-view data. We also show that the same framework can be extended to feed-forward 4D scene generation, broadening the task from static scenes to monocular dynamic inputs.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central idea is to convert the implicit 3D knowledge contained in camera-controlled video diffusion models into an explicit 3DGS decoder that can generate high-quality 3D representations. We instantiate this idea as a teacher–student framework: the video diffusion model acts as the teacher by producing RGB videos, and a 3DGS decoder operating in the same latent space acts as the student, as shown in Fig. 2. This section first reviews the video diffusion backbone used in our study (Sec. 3.1) and then describes how it supplies supervision for the 3DGS decoder (Sec. 3.2). We then present the 3DGS decoder design in Sec. 4 and describe the minimal modifications needed to apply the same pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised only by outputs from a camera-controlled video diffusion model, the approach depends on the teacher’s ability to generate view-consistent content. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed for this setting. We summarize the GEN3C components that are most relevant to our self-distillation framework.

Spatiotemporal 3D cache. GEN3C improves video consistency and camera-control accuracy by constructing a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos. Each $\mathbf{P}^{t,v}$ is a colored point cloud produced by unprojecting the depth estimate (Wang et al., 2024a) of an RGB image observed from camera viewpoint v at time t . The cache is arranged as an $L \times V$ array, where L denotes temporal length and V denotes the number of camera views.

Rendering and structured guidance. GEN3C uses this cache by rendering each point cloud from arbitrary camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane using camera pose $\mathbf{C}^t$ at time t . For a camera-pose sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache entries yields V videos of length L , which are stacked as image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify regions that the video diffusion model must complete. Together, the rendered images and masks provide structured visual guidance for generating camera-controlled videos.

Video variational autoencoder (VAE). Diffusion-based video generation systems commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space for efficient training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps it to a latent representation, i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The diffusion model is trained and evaluated in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered with the pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this work, we use the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where $\tau = 8$ is the temporal compression factor and $\sigma = 8$ is the spatial compression factor.

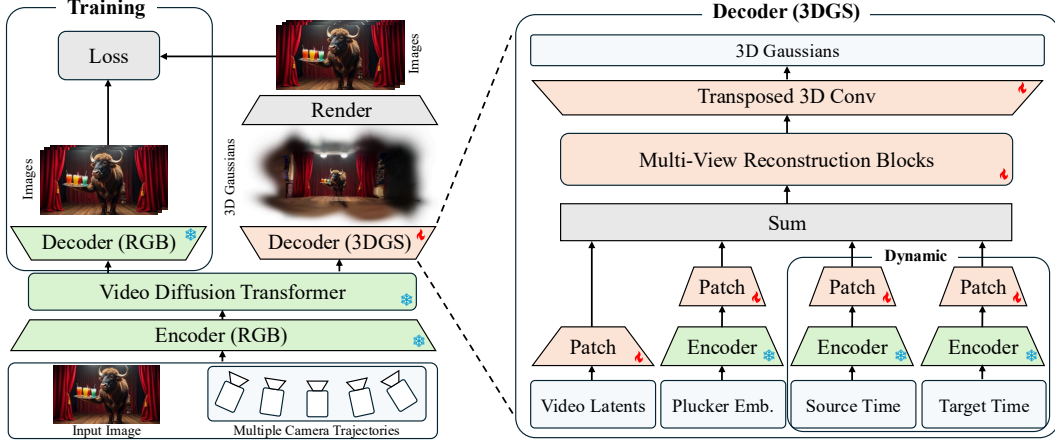


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds on a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train only a 3D Gaussian Splatting (3DGS) decoder by aligning renderings of generated 3DGS scenes with RGB-decoded generations from the frozen video model, while keeping the pre-trained autoencoder and diffusion model fixed. At inference time, the 3DGS decoder is used directly and no longer requires the RGB decoder. Time conditioning in the same decoder family allows the framework to extend from static to dynamic 3D scene generation.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student in a teacher–student paradigm where a camera-controlled video diffusion model $\mathcal{V}$ provides the supervision. To obtain broad synthetic coverage within this setup, we curate a large-scale set of diverse text prompts using large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand them into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. We decode $\mathbf{z}$ through two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ outputs video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, and the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from the 3DGS (*student*) are written as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match the RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ from the (*teacher*), closing the self-distillation loop.

Multi-trajectory supervision. To broaden viewpoint coverage around each input image, we sample $V = 6$ camera trajectories per input image, as illustrated in Fig. 3. For each trajectory v , we build a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ is trained to fuse the multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while completing disoccluded regions. With this self-distillation design, $\mathcal{D}_s$ learns from synthetic supervision supplied by $\mathcal{V}$, latent-space aggregation makes many generated trajectories tractable, and the explicit Gaussian output $\mathbf{G}$ supports view-consistent rendering for simulation and interactive use.

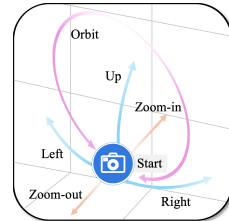


Figure 3: Six sampled camera trajectories used to broaden view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ maps synthesized multi-view latents from the video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We use 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit scene representation with fast rendering. The detailed decoder architecture is shown in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Many feed-forward reconstruction frameworks predict 3D Gaussians per input pixel, which restricts the number and resolution of views they can process. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a regime that exceeds the practical capacity of existing pixel-space approaches and GPU memory limits. The bottleneck is the attention mechanism over visual tokens, whose memory and compute grow with the number of pixels. We therefore avoid scaling reconstruction in pixel space and operate directly in the compressed video latent space produced by the camera-controlled video diffusion model. We denote the multi-view video latents as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the reconstruction blocks and predicts per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both represented in latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each component is first patchified to match the hidden dimension of the reconstruction blocks. No additional visual encoder is required; a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) converts video latents into flattened tokens for the reconstruction network. The summed inputs are processed by reconstruction blocks following Long-LRM (Ziwen et al., 2024), where each block contains one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice, resulting in 16 layers with hidden dimension 512. A transposed 3D convolution then maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to project these Plücker embeddings into the latent space. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins with $\mathcal{E}$, then concatenate the resulting latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so that it matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise the 3DGS decoder with image-based reconstruction losses so that rendered Gaussian views match the teacher video frames. The reconstruction objective combines a Mean Squared Error (MSE) loss $\mathcal{L}_{\text{mse}}$ with an LPIPS loss $\mathcal{L}_{\text{lpips}}$ (Zhang et al., 2018) computed using VGG (Simonyan & Zisserman, 2014) features.

Depth supervision. In our experiments, RGB-only supervision often led to flattened geometry. We therefore also supervise rendered depth maps using consistent video depth estimated by ViPe (Huang et al., 2025) and the scale-invariant depth loss $\mathcal{L}_{\text{depth}}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply L1 regularization to opacity through $\mathcal{L}_{\text{opacity}}$ and remove the Gaussians with the lowest 80% opacity, yielding a more compact representation.

Total loss. The full training objective is the weighted sum of these terms with weights λ_i :

$$\mathcal{L} = \lambda_{\text{mse}} \mathcal{L}_{\text{mse}} + \lambda_{\text{lpips}} \mathcal{L}_{\text{lpips}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{opacity}} \mathcal{L}_{\text{opacity}} \quad (1)$$

where we set $\lambda_{\text{mse}} = 1.0$, $\lambda_{\text{lpips}} = 0.5$, $\lambda_{\text{depth}} = 0.05$, and $\lambda_{\text{opacity}} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

The same self-distillation approach can be adapted to dynamic 3D scenes with limited architectural changes, as described below.

Self-distillation for dynamic scenes. The dynamic 3D setting follows the static design but replaces a single image input with a monocular video of length L and corresponding camera poses. The

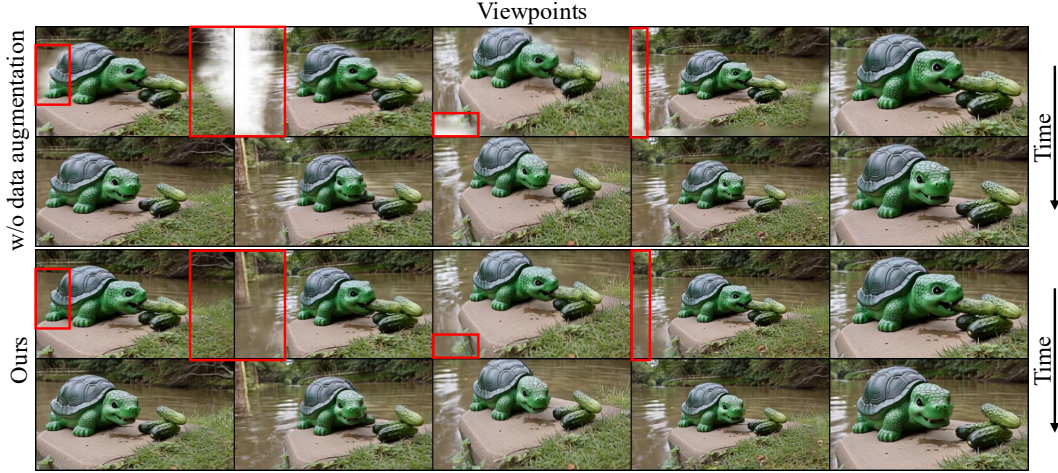


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder can produce artifacts in generated 3D Gaussians, especially low-opacity regions at early timesteps from extreme poses where supervision is sparse. We augment the supervision set with a motion-reversed video so that each timestep receives broader spatial coverage, reducing these early-timestep opacity artifacts.

video model then generates multi-view video latents that preserve the same underlying motion. For video inputs, we use a data-generation protocol analogous to the static case: diverse text prompts are sampled with large language models, videos are generated with video diffusion models (Cosmos, 2025; Wan, 2025), and the resulting videos are annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and define a dynamic 3DGS decoder $\mathcal{D}_s$ that generates time-dependent 3D Gaussians for all video frames. The dynamic decoder mirrors the static $\mathcal{D}_s$ architecture, except that its input is augmented with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned to each input frame, giving $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the raw target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated, augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, and encoded with the RGB encoder $\mathcal{E}$ into latent tensors $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target time step, decode the corresponding 3D Gaussians, and supervise them across all trajectories produced by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$, initializing the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zeros. The embeddings $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. For static 3D scenes, all frames from different timesteps can supervise the same 3DGS rendering because the scene does not change over time. For dynamic 3D scenes, the 3DGS is time-dependent, so only frames at the corresponding timestamp provide valid supervision. This narrower supervision can lead to a trivial solution in which the 3DGS for a given timestep ignores information from other timesteps (see Fig. 5).

To reduce this imbalance, we introduce dynamic data augmentation that improves supervision coverage across time. Early frames are typically associated with viewpoints near the input image, while later frames correspond to farther viewpoints. We therefore augment the training data with paired supervision views for each timestep, one near and one far. Concretely, we reverse the input video in frame order and pass it through the video model $\mathcal{V}$, producing six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six inward trajectories that move from far viewpoints toward the input. Combined with the original six outward trajectories, this provides 12 supervision views per timestep. This augmentation is used only during training: the reversed trajectories serve as additional supervision to prevent collapse from pruning artifacts and are not needed at inference time.

Table 1: **State-of-the-art comparisons.** We evaluate single-image-to-3D generation on RealEstate10K, DL3DV, and Tanks-and-Temples using the standard reported metrics.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** We show five rendered viewpoints from generated 3DGS scenes, illustrating novel-view synthesis across the evaluated examples.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. For training, we do not rely on existing captured multi-view datasets. Instead, we construct the *Lyra* dataset, where supervision for the 3DGS decoder is provided solely by our video model. The static 3D setup contains 59,031 images, and the dynamic 4D setup contains 7,378 videos. The data are generated from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both realistic and imaginative content. For each image (3D) or video (4D), we synthesize six camera trajectories, producing 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art methods for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, our main quantitative comparisons use the reported numbers from the corresponding papers.

Evaluation. Following prior work, we evaluate single-image-to-3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). We use the evaluation protocol from Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). Performance is measured with standard reconstruction metrics: PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports quantitative results for single image-to-3D generation across the evaluated benchmarks. Our method outperforms prior work on all listed benchmarks and metrics. Because our decoder learns from the camera-controlled video model, improvements in future video generative models should directly translate into stronger generated 3D reconstructions under this framework. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.



Figure 7: **Ablations.** We render the same extreme novel viewpoint after image-to-3DGS generation to compare design choices across the evaluated prompt distribution; corresponding depth visualizations and ablations are provided in Appendix C.4.

Qualitative results. We show novel-view renderings from generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. Across the shown static and dynamic examples, our method synthesizes plausible unseen regions while preserving consistency with the conditioning image or video. Additional qualitative comparisons on the *Lyra* dataset appear in Appendix C.

6.3 ABLATIONS

Tab. 2 studies the design choices that support generalization on diverse out-of-distribution prompts. We compare 3DGS renderings against videos generated by the camera-controlled video diffusion model used as supervision.

Real data only. As an alternative to self-distillation, we train a model using only real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets are common in previous reconstruction work, they provide narrower coverage than our generated prompt distribution and lead to weaker generalization on out-of-distribution scenes.

Self-distillation + real data. Joint training with self-distillation and real data does not improve over using self-distillation alone. This suggests that, for the evaluated setting, the synthetic teacher supervision is sufficiently diverse and consistent to train the reconstruction model without captured multi-view data.

No depth loss. Depth loss reduces flattened geometry and slightly improves image-based metrics. We visualize the corresponding depth maps in Appendix C.4.

No opacity pruning. We evaluate a variant that keeps all Gaussians rather than pruning them by opacity. Learning to remove low-opacity Gaussians makes the generated representation more compact and slightly improves visual quality. Rendering at $H = 704$, and $W = 1280$ takes 18ms with pruning compared with 30ms without pruning, yielding a $1.67\times$ speed up.

No LPIPS loss. Including the LPIPS loss improves robustness to input inconsistencies and better preserves high-frequency details.

No multi-view fusion. A simple alternative is to generate 3D Gaussians independently for each camera trajectory and then fuse them into a single point cloud. In contrast, learning multi-view fusion across trajectories with reconstruction blocks, where tokens attend to one another, yields clear advantages.

No Mamba-2. Replacing the joint Transformer/Mamba-2 design with Transformer-only blocks produces slightly lower quality. For one forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$, the joint blocks take 3213ms, whereas Transformer-only blocks take 20922ms, corresponding to a $6.5\times$ speedup.

Table 2: **Ablation study on the *Lyra* dataset.** The variants isolate data, loss, and architecture choices under the same evaluation setting.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

No latent-based 3DGS. Pixel-space approaches such as BTimer (Liang et al., 2025b) are limited to 12 input frames, whereas our latent-space decoder can use up to 726 input frames. Consequently, replacing latent-space 3DGS decoding with pixel-space decoding leads to out-of-memory in this high-view, high-resolution setting.

7 CONCLUSION

We presented *Lyra*, a feed-forward framework for 3D and 4D generation from a single image or video input. Rather than collecting captured multi-view datasets, the method uses camera-controlled video diffusion models as teachers in a self-distillation framework for a student 3DGS decoder. The decoder operates in the video model’s latent space and directly predicts 3D Gaussians without per-scene optimization or post-processing. This design improves coverage across diverse generated scenes within our evaluation setting, and the dynamic extension shows that the same principle can support 4D generation from monocular video. The scale and consistency of the generated scenes are currently bounded by the camera-controlled video diffusion model used as teacher. As these video models improve, the same framework should benefit in large-scale and view-consistent scene synthesis. Adapting auto-regressive techniques (Chen et al., 2024a) to this framework is a promising direction for broader environment generation. Finally, incorporating motion and tracking information within the reconstruction network, as explored in concurrent work (Lin et al., 2025), may further improve dynamic visual quality.

8 ETHICS STATEMENT

Like other generative AI technologies, our method could be misused to create misleading synthetic 3D or 4D content. At the same time, the intended contribution of this research is to support simulation, robotics, and embodied AI by providing scalable and controllable tools for generating environments under clearly specified inputs. We view the broadest positive impact of the work as arising from the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** The method provides controllable and physically consistent camera motion within generated 3D and 4D environments, supporting training and evaluation of agents that must perceive, navigate, and interact in complex scenes.
- **Improved scalability for data engines:** By reducing dependence on multi-view capture and per-scene optimization, the method can help create high-fidelity, customizable scenes at scale for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, this work offers a concrete example of how camera control, spatiotemporal generative models, and explicit 3D representations can be connected, which is relevant to computer vision, graphics, and embodied AI.

We caution against deceptive or wrongful uses of realistic generated media, especially cases where synthetic or unverifiable content could be presented as real. Safeguards such as provenance tracking, dataset documentation, and careful evaluation practices can help ensure that generative 3D and 4D models are developed and deployed in ways that benefit science and society.

9 REPRODUCIBILITY STATEMENT

To support reproducibility and future work on generative 3D and 4D scene reconstruction, we will release the training and inference code, model weights, and data for both settings upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section gives additional details about the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1, focusing on the refinement we use to improve its 3D consistency for our broader self-distillation pipeline.

Conservative mask refinement. One limitation of GEN3C’s forward warping is background leakage: object regions that are occluded in the source view, such as the lion body in Fig. 8, may become visible from a novel viewpoint but remain unmarked in the disocclusion mask. When this occurs, the video diffusion model receives incomplete foreground-completion guidance, as shown in Fig. 8 (second and fourth columns).

We address this issue with a conservative masking strategy. Because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume those regions may be occupied and derive refined disocclusion masks accordingly. This gives the video model guidance for potentially visible foreground areas that standard point-based rendering can miss.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This mesh approximates a surface boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ is the original disocclusion mask from forward warping and ϵ is a small tolerance. If the mesh surface is closer than the original points, we conservatively mask the corresponding region as a potential foreground disocclusion. This produces more reliable guidance for video generation under large viewpoint changes, as illustrated in Fig. 8 (third and fifth columns).

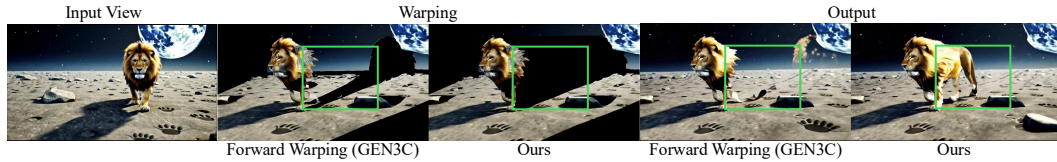


Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which uses forward-warped images for camera-control conditioning. We compare forward warping (left) with our conservative rendering function (right). The highlighted region is filled with background pixels by forward warping but is correctly masked by our conservative renderer, enabling foreground completion.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is computationally expensive, so we use the progressive training schedule in Tab. 3. We also initialize the dynamic model from the final static pre-trained model. Training takes 6 days on 8 NVIDIA A100 80 GB GPUs, and we use gsplat (Ye et al., 2025) for the 3DGS implementation.

Subsampling Gaussians. Pixel-aligned 3D Gaussian generation scales linearly with spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, which would produce 654,213,120 per-pixel Gaussians. We therefore generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based pruning further reduces this number to 2,044,416, yielding a more compact output representation.

Table 3: **Progressive training setup.** Static training increases resolution, temporal length, and view count before initializing dynamic training.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Joint Transformer-Mamba blocks. We use the Long-LRM (Ziwen et al., 2024) reconstruction block design, where each block consists of one Transformer layer followed by seven Mamba-2 layers. Repeating this block twice gives 16 layers with 512 hidden dimensions. Our network is substantially smaller than Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. Pure Transformers are often more expressive but are much slower than Mamba2 for training and inference. Since both configurations were trained under the same budget, we hypothesize that the Transformer-only variant would need many more iterations to match the rendering quality of the joint design.

Camera encoding. We encode cameras with Plücker embeddings, following prior work (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Because the RGB encoder expects 3 input channels while Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately. The resulting latent encodings are concatenated along the channel dimension.

Time encoding. To make the 1-dimensional time input compatible with the RGB encoder’s 3-channel input format, we concatenate a 2-dimensional sinusoidal embedding to the raw time value along the channel dimension. These values are then replicated over spatial dimensions before being encoded into the video latent space. This design lets us encode source and target times independently within the compressed latent representation.

Dynamic data augmentation. Fig. 9 visualizes the dynamic data augmentation procedure for two representative camera trajectories. The augmentation creates paired supervision views that observe the same motion state from different extreme viewpoints, improving spatial coverage at each timestep.

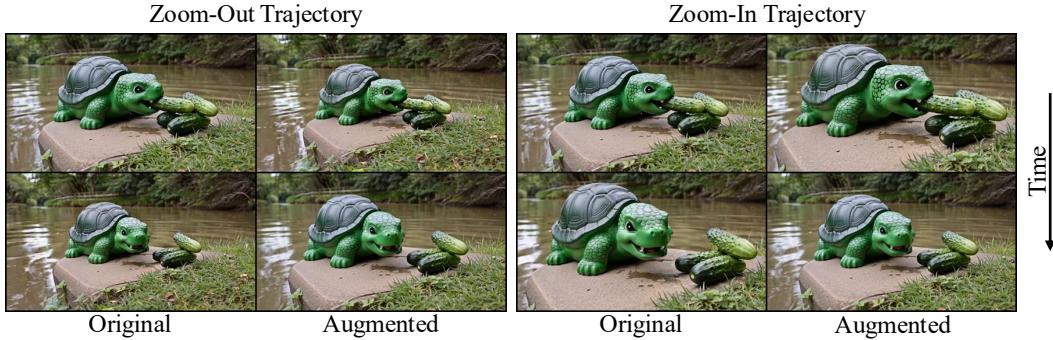


Figure 9: **Dynamic data augmentation videos.** We add motion-reversed videos to the supervision set so that each timestep is observed with broader spatial coverage, reducing low-opacity artifacts at early timesteps. The figure shows two example trajectories, zoom-out and zoom-in, together with their augmented videos whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) by rendering five views from generated scenes. In these evaluated examples, our method shows fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare with BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. We contacted the BTimer authors to run experiments on the same scenes as *Lyra*. Since BTimer is regression-based, we pair it with our GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as our model and provide RGB-decoded GEN3C videos as BTimer inputs. BTimer operates in pixel space and requires 12 input images. As discussed in the main paper, pixel-space models such as BTimer run out of memory when applied directly to the 726 high-resolution frames generated by GEN3C. We therefore uniformly subsample GEN3C frames. For static evaluation, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluation, we sample 11 frames spanning the viewpoint range and include the bullet-time frame corresponding to the motion being reconstructed, ensuring that the target motion state is present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 shows qualitative comparisons with BTimer (GEN3C), and Tab. 4a reports quantitative results. Our method produces higher visual quality with fewer artifacts in this evaluated setting.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimer (GEN3C), as described in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra* dataset. We crop videos to 512×512 for fair comparison because BTimer was primarily trained at that resolution. The results in Tab. 4b show that our method significantly outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth maps from generated 3DGS scenes. Depth supervision reduces flattened geometry while preserving visual quality.

Table 4: **Quantitative results on static and dynamic *Lyra* datasets.** The comparisons use the integrated BTimer (GEN3C) baseline described in the additional baseline details.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Static *Lyra* dataset comparison. (b) Dynamic *Lyra* dataset comparison.

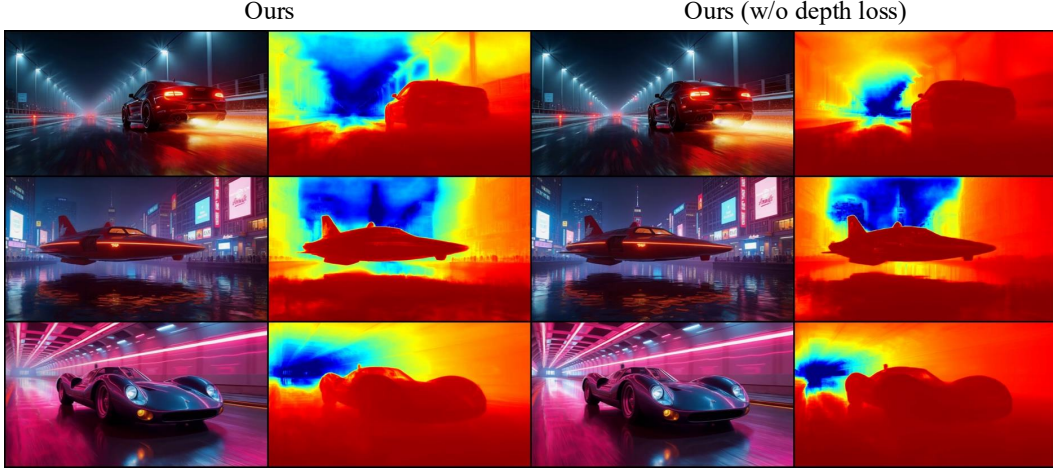


Figure 11: **Depth loss ablation.** We show 3DGS renderings and their depth maps. Adding depth supervision improves geometric structure by reducing flattened reconstructions while preserving visual quality.

D ROBOT SIMULATION

A major motivation for diverse 3D generation is the ability to synthesize simulation environments for autonomous agents. We therefore build an end-to-end export pipeline: first, our model generates 3D scenes from text; next, the 3D Gaussians are exported as a .ply file; finally, the .ply files are converted into .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function for .usdz creation. The resulting file can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. This allows AI-based robots to be trained and tested across varied generated conditions. Early demo results will be released upon acceptance, and Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section provides additional related work and positions our approach relative to broader 3D and 4D generation literature. Fig. 13 highlights differences between our framework and prior systems such as CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early 3D generation methods often focused on single object categories, extending GANs into 3D by using neural renderers as inductive biases (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). CLIP-based supervision (Radford et al., 2021) broadened this area toward more flexible asset creation, including text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To improve structural coherence, many methods enforce multi-view consistency by generating scenes from multiple perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a;

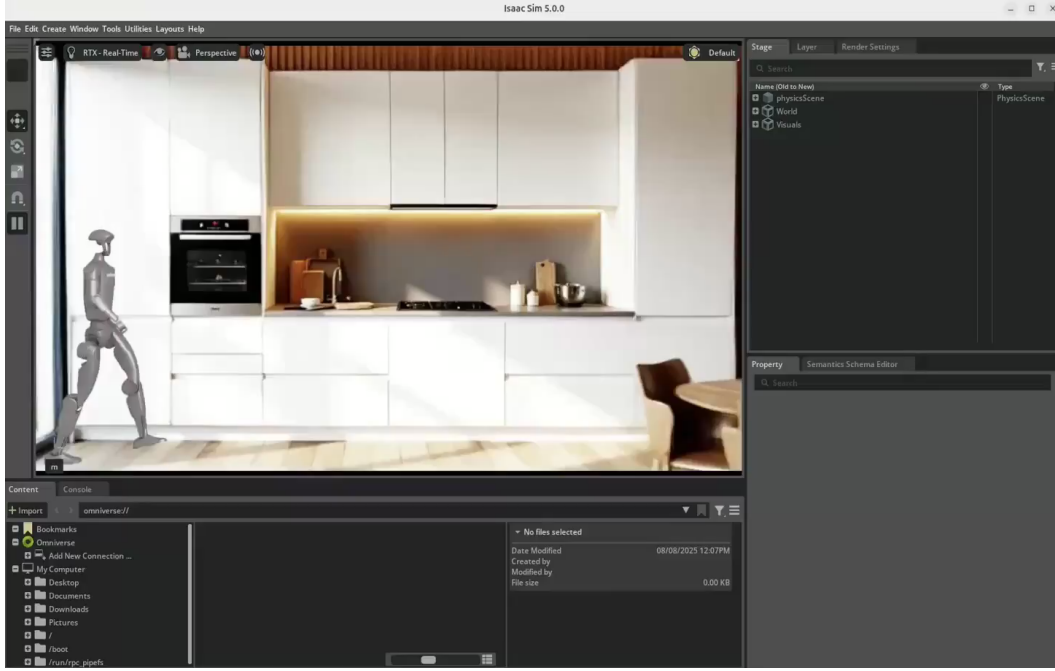


Figure 12: **Robot Simulation.** We show a frame from Isaac Sim 5.0 using a generated 3DGS scene from our method as the input environment, illustrating one downstream simulation scenario enabled by explicit 3D output.

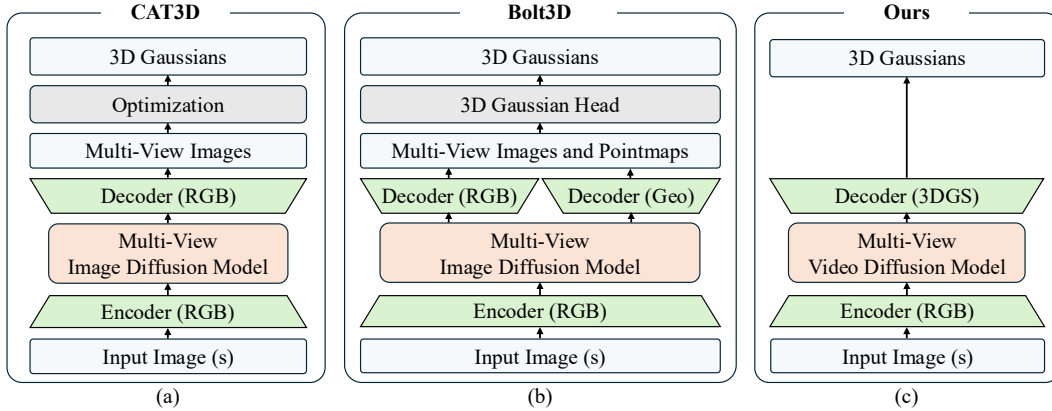


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) generates novel-view images with a multi-view image diffusion model and then reconstructs 3D Gaussians through optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps with a geometry autoencoder and predicts the 3D scene using a feed-forward 3D Gaussian head instead of optimization. Our framework uses a pre-trained camera-controlled video diffusion model and *directly* decodes multi-view video latents into 3D Gaussians, extending the scope from generated views to explicit scene representations.

Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others synthesize scenes through iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A related line lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations combined with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary work studies feed-forward methods that directly predict 3D content from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Unlike our method, these approaches produce static 3D scenes. Other methods specialize in particular scene types, such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but does not target diverse generated scenes or large viewpoint changes.

4D generation. 4D generation, or dynamic 3D scene synthesis, has progressed rapidly. Many methods use text prompts or images as guidance. After large-scale generative models were introduced for this task (Singer et al., 2023), subsequent systems improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, many approaches lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other work adds physics priors to generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), improves motion controllability with template-driven methods (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or studies compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another line extends 3D GANs to 4D by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). These methods are often constrained by small-scale, single-category datasets, which limits generalization. Many are also object-centric and de-emphasize background content. Their visual fidelity therefore remains below that of recent video generation models, whose priors our method leverages.

Camera-conditioned video models. Recent work has increasingly incorporated camera control into video diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) improved conditioning with Plücker-coordinate camera representations, while others follow MotionCtrl by using flattened poses (Wang et al., 2025b). Another line (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) controls camera motion without adding trainable parameters. Most of these methods rely on U-Net-based architectures, and other approaches explore low-rank camera-control adaptations (Zhang et al., 2025b).

Several recent methods continue extending U-Net-based camera-control systems (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others explore diffusion transformers. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but uses predefined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) study pose and geometry estimation with video DiTs. CAT4D (Wu et al., 2025b) fine-tunes a multi-view backbone into a multi-view video diffusion model. Large-scale synthetic data further improves performance, as shown by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong camera-controlled video generation with transformer architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel-view synthesis from a single input image. A related direction edits camera movement in videos with underlying motion through camera retargeting (Seo et al., 2025; Ma et al., 2025). Another widely used strategy is to condition camera-controlled video generation on a spatiotemporal cache, such as point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).